

# Designing Technology to Support Awareness, Self-Help, and Create Safe Spaces for Women Affected by Forced Marriages

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## Abstract

This research explores how interactive technologies can raise awareness, support self-help, and create safe spaces for women affected by forced marriages. Despite legal protections, FMs persist globally, including in Switzerland, where patriarchal family structures contrast with egalitarian societal norms. The study is structured into three stages: (1) supporting awareness and realization of abuse, (2) assisting in help-seeking through self-help and enhanced communication with support organizations, and (3) facilitating life repair to ensure safety and independence post-FM. Collaborating with the Swiss Competence Center against Forced Marriages, this research uses participatory, feminist, and trauma-informed approaches. The expected outcomes include tailored technological solutions for the unique challenges of FMs, offering insights into designing for complex social, cultural, and familial dynamics. This doctoral research aims to contribute to HCI by demonstrating how technology can empower individuals in patriarchal contexts within Western societies.

## CCS Concepts

• Human-centered computing → Empirical studies in HCI.

## Keywords

Forced Marriage, Martial Captivity, Domestic Abuse, Intimate Partner Violence, Women's Safety

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## 1 Research Status

I am a Fast Track PhD student at the University of Zurich, working under the supervision of Professor Elaine M. Huang at the People and Computing Lab. I began this combined MSc-PhD program in 2021, which is designed for high-achieving students and accelerates both Master's and PhD studies. Having completed my coursework

last year, I am now midway through my PhD journey, with an expected completion date of Mid-to-late 2026. By November 2024, I will have defended my PhD proposal.

The CHI 2025 Doctoral Consortium will be my first such experience, offering a valuable opportunity to engage with external advisors and peers at this stage of my PhD. As I refine my research and prepare for the dissertation's final phases, I am eager to gain insights from the broader HCI community, which will enhance the practical impact of my work. My research aims to advocate for marginalized communities affected by FM, supporting organizations like CoCFM through ethical, sensitive, and participatory research that gives affected individuals agency over the technologies designed for them. In terms of my long-term career goals, I am enthusiastic about staying in academia. After completing my PhD, I plan to pursue a postdoctoral position to further explore the intersection of culture, feminism, and technology. This academic path would allow me to deepen my research, contribute to new advancements, and continue fostering real-world impact for vulnerable communities.

## 2 Introduction

Forced marriage (FM) remains a critical human rights issue globally, despite legal protections and international efforts to prevent it. The Universal Declaration of Human Rights [17] emphasizes that marriage must be based on free and full consent, underscoring the right to choose one's life partner. Unlike arranged marriages, forced marriages coerce individuals—most often young women—into unions against their will, violating their rights to self-determination, bodily integrity, and freedom of movement [9, 18].

This research focuses on FM in Switzerland, in collaboration with the Swiss Competence Center against Forced Marriages (CoCFM). In 2022, 344 cases of FM were reported in Switzerland, with the real number likely much higher [4], reflecting a global issue affecting an estimated 15.4 million people [10]. FM poses unique challenges compared to domestic violence (DV) and intimate partner violence (IPV) due to its entanglement with family, community, and patriarchal traditions, which create significant barriers to help-seeking.

My PhD research addresses how technology can support FM survivors, whose needs differ from those affected by DV and IPV. The complex social structures of FM require tools that account for the influence of extended families and cultural pressures. Using participatory, feminist, and trauma-informed design approaches, I aim to explore how technology can raise awareness, facilitate self-help, and create safe spaces for survivors. Collaborating with CoCFM and other relevant organizations, I will employ qualitative

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methods to understand survivors' lived experiences and develop technology-driven interventions.

### 3 Related Work

In HCI, significant research has explored technologies addressing DV and IPV, which share some elements with FM. However, FM poses unique challenges due to its entanglement with family, community, and patriarchal traditions [14], which make help-seeking more complex. FM is often ingrained in tradition from childhood, making it distinct in both its dynamics and implications. Despite these differences, research on DV and IPV can inform technology design for FM.

For instance, Tarzia et al. [16] explored young women's perceptions of technologies supporting domestic violence survivors, highlighting the potential of technology to offer accessible help, raise awareness, and protect safety. Similarly, Glass et al. [6] developed MyPlan, an app that educates women about IPV and provides guidance for supporting affected individuals. Their evaluation showed a decrease in physical and sexual abuse after 12 months of use [7]. Initially focused on gender and power, feminist approaches have since expanded to include intersections of gender with race, class, national origin, sexual orientation, age, and disability [5]. This has led to the development of culturally sensitive frameworks, such as Rabaan & Dombrowski's Islamic Feminist framework [12] and Sambasivan et al.'s South Asian Feminist methodology [13], emphasizing the importance of cultural context in designing interventions.

Further research has examined technology design in patriarchal environments, such as shared device usage [1]. Sultana et al. [15] stressed the importance of designing within patriarchal constraints, while Younas et al. [19] showed how Facebook groups offer anonymous support for sensitive issues in these communities. Other studies explored peer-support mechanisms in religious contexts [11], while Bellini et al. [2, 3] developed ChoicePoint, a tool fostering empathy in abusive contexts.

While this body of research has advanced DV and IPV interventions, there remains a gap in designing technologies for FM, particularly within Western contexts. FM survivors face dual pressures from their families and the egalitarian norms of society. My research addresses this gap by developing tailored technological solutions that account for the complex cultural, social, and familial dynamics that define FM.

### 4 Research Questions

The goal of this PhD thesis is to explore how technology can raise awareness, support self-help, and create safe spaces for women affected by FMs in Switzerland. This research is structured around three key stages: (i) **Awareness and Realization of Abuse**, (ii) **Active Help-Seeking**, and (iii) **Life Repair Post-FM**.

The first stage (i) investigates how technology can support awareness and realization of abuse among individuals who may not initially recognize their situation as abusive. The second stage (ii) examines how technology can assist those currently facing FMs in taking action, whether through self-help strategies or by enhancing communication and support from organizations. The third stage (iii) explores how technology can facilitate life repair and provide ongoing support for women who have left forced marriages, ensuring

their safety, well-being, and successful transition to independence. To achieve this goal, the following research questions aligned with these stages will be examined in my dissertation:

**RQ1:** How does technology contribute to creating awareness and supporting the realization of abuse among individuals affected by gender-based violence (GBV) (e.g. FM), and how does it influence their pathways to seeking help?

**RQ2:** What are the challenges and opportunities for technology in the context of forced marriages, and how can these insights inform directions for technological design interventions?

**RQ3a:** How can technology improve communication with support organizations, making it safer and more effective for individuals affected by forced marriages who are seeking help?

**RQ3b:** How can technology facilitate self-help strategies for individuals affected by forced marriages, empowering them to seek help independently?

**RQ4:** How can we design technology to create safe spaces and support life repair for women who have left forced marriages, fostering safety, well-being, and a transition to independence?

### 5 Research Approach

To address my research questions, I adopt a human-centered design (HCD) approach, focusing on the design, study, and evaluation of technologies in collaboration with the CoCFM. Given the sensitivity of FM, I primarily use qualitative methods, including semi-structured interviews, participatory design interviews, workshops, and focus groups. These methods allow for in-depth insights into the complex and often traumatic experiences of those affected, which would be difficult to capture through quantitative approaches.

Ethical and trauma-informed practices are essential throughout the research. To ensure participants' safety and anonymity, I collect minimal personal data, using pseudonyms and avoiding demographic details or identifiable information. Additionally, I have engaged in educational workshops with CoCFM to ensure that the research is conducted in a safe, culturally sensitive, and respectful manner. Inspired by feminist HCI values, my approach is designed to respect participants' experiences and provide a supportive environment for sharing. All our studies are reviewed by the University's Ethics Committee and have also been reviewed by the legal department of CoCFM, ensuring compliance with ethical and legal standards.

The research is structured into five interconnected studies, each aligned with a specific RQ and one of the three stages defined in this research. Study 1 addresses RQ1 and focuses on the **Awareness and Realization of Abuse Stage**. It is a mixed-methods study combining an online survey and qualitative interviews to explore how individuals affected by FM come to recognize their situation as abusive. This study will also generate design implications and best practices for organizations like CoCFM.

Study 2 addresses RQ2 and spans all stages. It involves participatory design interviews with CoCFM advisors and affected individuals to understand how they envision technology addressing FM at

the individual, community, and societal levels. The insights from this study will inform design implications and future interventions.

Study 3a addresses RQ3a and focuses on the **Active Help-Seeking Stage**. It aims to improve communication between individuals and support organizations through the development of a case management tool, which will be tested for its impact on advisor efficiency and user safety. Study 3b, also within the **Active Help-Seeking Stage**, addresses RQ3b and focuses on developing a self-help web application to empower individuals affected by FM to make independent decisions and access resources securely, with particular attention to designing for discretion and privacy.

Finally, Study 4 addresses RQ4 and focuses on the **Life Repair Post-FM Stage**. It involves participatory design interviews to explore how safe spaces—both physical and digital—can be created for survivors of FM, fostering safety and supporting their transition to independence.

## 6 Research to Date

### 6.1 Study 2: An Investigation of the Practice of Forced Marriage and the Potential for Design Interventions

This study addresses RQ2 and spans all three key stages. The goal of this study was to understand the systemic nature of FMs, the unique challenges faced by those attempting to escape or resist FM, and how technology could support them. By exploring these factors, I aimed to identify opportunities and challenges for technological interventions.<sup>1</sup>

Using a qualitative approach, I conducted semi-structured participatory design interviews and workshops with five CoCFM advisors and nine individuals affected by FM. The interviews explored participants' experiences with FM, their help-seeking strategies, and their interactions with technology. In the second part of the interviews, participants were encouraged to brainstorm and sketch potential technological solutions that could help them navigate or escape these situations. This part uncovered how technology could be leveraged at the individual, community, and societal levels to address these challenges.

The findings revealed that FM is not an isolated event but a systemic issue deeply embedded in societal norms, with lifelong effects. Participants frequently lacked support from their communities, and external systems often failed to grasp the complexity of their situations. First, technology was frequently misused by perpetrators for surveillance and control, raising significant privacy and security concerns. This highlighted the need for discreet technological interventions that assume limited privacy, provide secure access, and do not rely on easily detectable mechanisms, such as passwords.

Second, participants emphasized the need for culturally relevant and accessible information. Many were unaware of their rights or available resources, underscoring the importance of providing tailored guidance that resonates with their lived experiences. Technology must not only educate but also offer culturally sensitive support that respects their preferences, including non-legal solutions or alternative forms of justice.

Third, family and community pressures heavily influenced help-seeking behavior. Participants expressed that transitioning from a patriarchal environment to an egalitarian society was particularly challenging. As such, technology should guide users through this transition, supporting their newfound autonomy and helping them adapt to egalitarian values while managing the associated cultural and emotional complexities.

## 7 Research in Progress & Proposed

### 7.1 Study 1: Examining Technology's Role in Recognizing and Shaping Help-Seeking Pathways

This study aims to address RQ1 and **Awareness and Realization of Abuse Stage**, by exploring the role of digital platforms in raising awareness of abuse among individuals affected by GBV. Prior research highlights the potential of digital platforms for trauma unmuting and community building [11, 15, 19], but little is known about their impact on the recognition of abuse and subsequent help-seeking behavior [8]. We aim to fill this gap by investigating how digital content influences these processes.

Study 1 is a mixed-methods study, combining an online survey with semi-structured interviews to explore participants' interactions with GBV-related digital content. We have completed the survey, which gathered 66 full responses and over 100 partially filled ones. Given the sensitivity of the topic, many questions were optional, allowing participants to share only what they felt comfortable with. We are currently in the process of analyzing this survey data.

In addition to the survey, we have conducted five in-depth qualitative interviews so far. These interviews explore how specific digital content helped participants recognize their situations as abusive and how this recognition influenced their decisions to seek help.

### 7.2 Study 3a: Improving Communication with Support Organizations

Effective communication with support organizations is critical for individuals seeking help. Findings from RQ2 revealed several challenges, especially with organizations like CoCFM, which often rely on manual note-taking and have limited advisor availability. Affected individuals frequently have restricted phone access, making it crucial for advisors to quickly retrieve case information during emergency calls or unscheduled sessions. This study aims to answer RQ3a and focuses on the **Active Help-Seeking Stage**, by developing a case management system that allows advisors to access relevant case details quickly and efficiently.

We have completed the design and development phase of the case management tool through iterative design and feedback sessions with CoCFM. The next phase involves a two-week evaluation, where advisors will use the tool in practice with low-risk cases who have unrestricted phone access. During this period, we will gather feedback through daily and weekly surveys to assess the tool's impact on advisor efficiency and communication effectiveness.

One of the key challenges in this phase is ensuring the safe and ethical use of the system in real-life scenarios. Testing the tool in a

<sup>1</sup>This study was published at NordiCHI24

real-world setting, where the risk to individuals must be minimized, requires careful planning, and I welcome feedback and advice on how to navigate this challenge effectively.

### 7.3 Study 3b: Improving Self-Help Technologies

The focus of RQ3b is on improving self-help technologies for individuals affected by FM, and therefore on the **Active Help-Seeking Stage**. This study aims to provide safer, independent options for those who may prefer not to rely on support organizations like CoCFM, offering tools that allow them to explore pathways and make informed decisions that align with their unique circumstances. By taking an intersectional feminist viewpoint, we aim to empower individuals by providing resources that support self-directed choices, whether they seek assistance from community leaders or explore other resolutions.

Building on insights from RQ2, we plan to begin the design phase of a self-help web application in November 2024. The app will be accessible from any device without requiring downloads, ensuring discretion. It will provide resources such as educational materials, legal guidance, interactive decision-making guides, and safety planning tools, enabling users to gather information privately and securely, without needing to make risky calls to support organizations. Once the design and development are complete, we will begin a 1-2 month evaluation phase in collaboration with CoCFM, testing the tool with low-risk individuals who have private access to their phones. During this time, we will collect qualitative feedback to understand how the tool affects users' decision-making and help-seeking behaviors.

The key challenge, as with Study 3a, is ensuring the ethical testing of the tool in real-world scenarios without exposing individuals to risk. Although we will work with low-risk participants deemed safe by CoCFM, additional measures to ensure safety during testing will be critical, and we welcome feedback on how to approach this challenge.

### 7.4 Study 4: Designing Safe Spaces for Life Repair Post-FM

The focus of RQ4 is on the **Life Repair Post-FM Stage**, where individuals who have left forced marriages seek to rebuild their lives and transition to independence. This stage is crucial as it involves not only continued help-seeking but also the creation of safe spaces that support long-term recovery and well-being. The goal of this study is to provide design implications for better support mechanisms that cater to the diverse needs of women navigating life after FM.

The study will primarily involve conducting in-depth interviews. These interviews will explore how, if at all, technology has served as a safe space for individuals post-FM, as well as the barriers for those who have not utilized such resources. By gathering insights into participants' experiences—both positive and negative—this study aims to identify the requirements for digital spaces to be considered safe and supportive during the recovery process.

While this study will inform the design of digital safe spaces, implementing and testing such spaces is beyond the scope of this dissertation. Due to the sensitive nature of the population and recruitment challenges in Switzerland, the conclusions of this study

will focus on generating guidelines for future research and design in this area. I am aware that this study opens new directions for exploration rather than providing definitive solutions. Feedback from the consortium on how best to approach this aspect is welcome.

## 8 Conclusion

My research is dedicated to developing and understanding digital interventions for individuals affected by forced marriage, a complex issue deeply rooted in cultural, social, and gender dynamics. By leveraging technology, my dissertation aims to contribute in several significant ways:

- **Enhanced Awareness:** Strategies that use digital content to increase recognition of abusive scenarios within gender-based violence.
- **Interventions:** Development and real-world testing of self-help technologies and communication tools based on insights from in-depth interviews with affected individuals and advisors.
- **Ethical Engagement:** Guidelines for ethically engaging at-risk populations, ensuring interventions respect their unique needs and experiences.
- **Connecting Theory and Practice:** Bridging theoretical insights with practical applications to enhance the efficacy of NGO interventions in real-world contexts.

These contributions aim to deepen our understanding of the intersection between technology, culture, and gender-based violence, particularly in Western settings where such issues are often underexplored in HCI. By focusing on these areas, my research strives to empower marginalized populations and enhance their interaction with digital tools in meaningful ways, fostering autonomy, safety, and well-being.

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